Program 1

Build a Docker Container from a Custom Dockerfile

Program Structure:

program1/
--------->Dockerfile
--------->app.py
--------->requirements.txt

Procedure:

step 1: create the dockerfile and the following code

1RV24MC073_pavithra@pavithra-HP-Laptop-15-db1xxx: ~/program1

GNU nano 6.2 Dockerfile

```
# Use a slim Python base image
FROM python:3.9-slim

# Set working directory inside the container
WORKDIR /app

# Copy the requirements file and install dependencies
COPY requirements.txt .
RUN pip install --no-cache-dir -r requirements.txt

# Copy the application code
COPY . .

# Set Flask environment variables
ENV FLASK_APP=app.py
ENV FLASK_RUN_HOST=0.0.0.0
ENV FLASK_RUN_PORT=5000

# Expose the port Flask runs on
EXPOSE 5000

# Command to run the Flask application
CMD ["flask", "run"]
```

step 2: create python application file
nano app.py

```
1RV24MC073_pavithra@pavithra-HP-Laptop-15-db1xxx: ~/program1
  GNU nano 6.2                                          app.py
from flask import Flask

app = Flask(__name__)

@app.route("/")
def hello():
    return "Hello, Docker!"

if __name__ == "__main__":
    app.run(host="0.0.0.0", port=5000)
```

step 3: create requirements.txt file

nano requirements.txt

```
1RV24MC073_pavithra@pavithra-HP-Laptop-15-db1xxx: ~/program1
  GNU nano 6.2                                  requirements.txt
flask
```

step 4: build docker image

sudo docker build -t program1 .

```
1RV24MC073_pavithra@pavithra-HP-Laptop-15-db1xxx: ~/program1
bash: /home/pavithra/dev/flutter/bin/cache: No such file or directory
1RV24MC073_pavithra@pavithra-HP-Laptop-15-db1xxx:~$ mkdir program1
mkdir: cannot create directory ‘program1’: File exists
1RV24MC073_pavithra@pavithra-HP-Laptop-15-db1xxx:~$ cd program1
1RV24MC073_pavithra@pavithra-HP-Laptop-15-db1xxx:~/program1$ ls
1RV24MC073_pavithra@pavithra-HP-Laptop-15-db1xxx:~/program1$ nano Dockerfile
1RV24MC073_pavithra@pavithra-HP-Laptop-15-db1xxx:~/program1$ nano app.py
1RV24MC073_pavithra@pavithra-HP-Laptop-15-db1xxx:~/program1$ nano requirements.txt
1RV24MC073_pavithra@pavithra-HP-Laptop-15-db1xxx:~/program1$ docker build -t program1 .
[+] Building 3.6s (11/11) FINISHED                                          docker:default
 => [internal] load build definition from Dockerfile                                  0.0s
 => => transferring dockerfile: 531B                                                  0.0s
 => [internal] load metadata for docker.io/library/python:3.9-slim                    3.2s
 => [auth] library/python:pull token for registry-1.docker.io                         0.0s
 => [internal] load .dockerignore                                                     0.0s
 => => transferring context: 2B                                                       0.0s
 => [1/5] FROM docker.io/library/python:3.9-slim@sha256:2d97f6910b16bd338d3060f261f53f144965f755599aab1acda1e13cf17  0.0s
 => [internal] load build context                                                     0.0s
 => => transferring context: 791B                                                     0.0s
 => CACHED [2/5] WORKDIR /app                                                         0.0s
 => CACHED [3/5] COPY requirements.txt .                                              0.0s
 => CACHED [4/5] RUN pip install --no-cache-dir -r requirements.txt                   0.0s
 => [5/5] COPY . .                                                                    0.1s
 => exporting to image                                                                0.1s
 => => exporting layers                                                               0.0s
 => => writing image sha256:a60a804f9b59fbbbe328a342e209ed3b42f9504730e37a5d80906d8a2132054f  0.0s
 => => naming to docker.io/library/program1                                           0.0s
```

step 5: run the container

sudo docker run -p 5000:5000 program1

```
1RV24MC073_pavithra@pavithra-HP-Laptop-15-db1xxx:~/program1$ docker run -d -p 5000:5000 --name flask-container program1
89e4e5d8de55fcdd81fddc73cb0caaee37bc1365c4a3a3a8489d38043261f20f
1RV24MC073_pavithra@pavithra-HP-Laptop-15-db1xxx:~/program1$
```

Step 6 : veryify the application

open your browser and go to http://localhost:5000